CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 21: Trigonometrical Identities

Previous year board questions, topic-wise: 13 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 The three basic identities

1. Assertion (A): If $\sin^2 A + \sin A = 1$, then $\cos^4 A + \cos^2 A = 1$. Reason (R): $1 - \sin^2 A = \cos^2 A$. (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A ICSE 2025, 1 mark
2. Statement 1: $\sin^2\theta + \cos^2\theta = 1$. Statement 2: $\operatorname{cosec}^2\theta + \cot^2\theta = 1$. Which is valid? (a) only 1 (b) only 2 (c) both (d) neither ICSE 2024, 1 mark
3. $(1 + \sin A)(1 - \sin A)$ equals: (a) $\operatorname{cosec}^2 A$ (b) $\sin^2 A$ (c) $\sec^2 A$ (d) $\cos^2 A$ ICSE 2023, 1 mark

2 Proving identities

4. Prove that $\dfrac{(\cot A + \tan A - 1)(\sin A + \cos A)}{\sin^3 A + \cos^3 A} = \sec A \operatorname{cosec} A$. ICSE 2025, 4 marks
5. Factorise $\sin^3\theta + \cos^3\theta$. Hence prove that $\dfrac{\sin^3\theta + \cos^3\theta}{\sin\theta + \cos\theta} + \sin\theta\cos\theta = 1$. ICSE 2024, 4 marks
6. Prove that $(\sin^2\theta - 1)(\tan^2\theta + 1) + 1 = 0$. ICSE 2023, 4 marks
7. Prove that $\dfrac{\sin A}{1 + \cot A} - \dfrac{\cos A}{1 + \tan A} = \sin A - \cos A$. ICSE 2020, 3 marks
8. Prove that $\left(\dfrac{1 - \tan\theta}{1 - \cot\theta}\right)^2 = \tan^2\theta$. ICSE 2020, 4 marks
9. Prove that $(\operatorname{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta) = 1$. ICSE 2019, 3 marks
10. Prove that $\sqrt{\sec^2\theta + \operatorname{cosec}^2\theta} = \tan\theta + \cot\theta$. ICSE 2018, 3 marks
11. Prove that $(1 + \cot\theta - \operatorname{cosec}\theta)(1 + \tan\theta + \sec\theta) = 2$. ICSE 2018, 4 marks
12. Prove that $\dfrac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$. ICSE 2017, 3 marks

3 Complementary angles

13. Without using tables, evaluate $\sin^2 28^\circ + \sin^2 62^\circ + \tan^2 38^\circ - \cot^2 52^\circ + \dfrac{1}{4}\sec^2 30^\circ$. ICSE 2017, 3 marks

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